

MSC Project

Project Proposal

Student Name: Abhishek Maheshkumar Parikh

Student ID: A00075992

Title

Multi-Modal Waste Characterisation for High-Quality Refuse-Derived Fuel Production Using Machine Learning

Introduction

The increasing amount of household and industrial waste has created significant environmental and waste management challenges worldwide. One approach to reducing landfill waste is the production of Refuse-Derived Fuel (RDF), where suitable waste materials are processed and used as an alternative energy source. However, identifying and sorting waste materials suitable for RDF production can be time consuming and may require expert knowledge.

Recent advances in machine learning and computer vision have shown promising results in automating waste classification tasks. Most existing studies focus on image based waste classification, while relatively few combine image information with material related characteristics that influence RDF quality. This project aims to investigate whether a multi-modal machine learning approach can improve waste characterisation and support RDF suitability assessment.

Problem Statement

Current waste sorting and RDF assessment processes often depend on manual inspection, which can be time consuming, expensive and unpredictable. Many existing machine learning studies focus only on image based waste classification and do not consider additional

material characteristics that influence RDF quality. As a result, there is a need for a system that can classify waste materials and predict RDF suitability using both visual and material related data.

Research Questions

- Can machine learning accurately classify waste materials from images?
- Can material-related features improve RDF suitability prediction?
- Does a multi-modal approach perform better than image-only classification?

Aim

To develop and evaluate a machine learning based multi-modal system for waste classification and RDF suitability assessment.

Research Objectives

- Review literature on waste classification and RDF production.
- Identify suitable datasets for waste classification.
- Develop a CNN-based waste classification model.
- Develop an RDF suitability prediction model using material-related features.
- Compare different machine learning algorithms.
- Evaluate model performance using standard metrics.
- Develop a Python-based prototype.

Scope of the Research

This project will focus on the design, implementation, and evaluation of a prototype system for waste characterisation. The study will utilise waste images and selected material related features such as material type, moisture content, contamination level, combustibility, and estimated calorific value.

The machine learning component will include image classification techniques and predictive modelling approaches to assess RDF suitability. The prototype will be developed using Python and relevant machine learning libraries.

The project is limited to experimental evaluation using publicly available or simulated datasets and will not involve deployment in a real industrial environment. Economic analysis, large scale operational deployment, and hardware-based waste sorting systems are outside the scope of this research.

Proposed Algorithms:

Image Classification:

- CNN (Convolutional Neural Networks)
- MobileNetV2
- ResNet50

RDF Prediction:

- Random Forest
- XGBoost

Legal, Social, Ethical and Professional Considerations

The project will primarily use publicly available datasets and will not involve personal or sensitive information. Ethical considerations include ensuring transparency in machine learning predictions and avoiding misleading performance claims. Socially, the research may contribute to improved waste management and sustainability practices. Professional standards relating to research integrity, proper referencing and responsible AI use will be followed throughout the project.

Background

Literature review covering:

- Waste management challenges.
- Refuse-Derived Fuel (RDF) production.
- Waste classification using computer vision.
- Convolutional Neural Networks (CNNs).
- Transfer learning approaches such as MobileNetV2 and ResNet50.
- Multi-modal machine learning.
- RDF quality assessment factors.
- Existing datasets such as TrashNet and TACO.
- Current limitations in the literature and research gap.

Proposed Datasets:

- TrashNet Dataset
- TACO Dataset

Potential RDF Features:

- Material Type
- Moisture Content
- Contamination Level
- Combustibility
- Calorific Value